PHY 317 – Classical Mechanics – Fall 2026

Course/contact information

Hours: Mon 1:40–2:55; Wed, Fri 1:20–2:35
Room: Sabin-Reed 308
Credits: 4
Text: Taylor, *Classical Mechanics* (2005)
Prerequisites: PHY 210 and PHY 215

Instructor: Michael Lerner
Email: mglerner@smith.edu
Office: McConnell 303
Website: Moodle
Python: `https://posit.smith.edu`

Office hours: **TO BE SET** after the week-1 scheduling poll. I also have an open-door policy: stop in whenever my door is open, which is most of the time.

Course structure

PHY 317 is Smith's upper-level mechanics course. It builds directly on PHY 117 (and a little of PHY 118) and uses the mathematics from PHY 210 constantly. We follow Taylor: Newtonian mechanics with realistic forces (Ch. 1–4), oscillations (Ch. 5), the calculus of variations and Lagrangian mechanics (Ch. 6–7), central forces and orbits (Ch. 8), non-inertial frames (Ch. 9), coupled oscillators and normal modes (Ch. 11), and chaos (Ch. 12).

The course meets three times a week for 75 minutes, with a lot of group work at the board. There are short pre-class check-ins most days, weekly homework, three in-class exams, and a final exam with redemption problems. The grade breakdown is below. Read the assigned sections *before* class; class time assumes you have.

Just-in-time review. The course calendar on Moodle has a "Prerequisites" column that lists, for each new topic, the earlier material it leans on (by Knight chapter for intro physics and by Felder & Felder chapter for math methods). There are two good ways to use it: *just-in-time* – review those concepts before we get to the topic – or *just-after-time* – if a new idea is rough going, use the list to find and shore up the foundation right after class. (This framing, and the column itself, are Will Raven's, from his Fall 2025 version of this course.) If you find the prerequisite material itself is the problem, come see me early; closing a gap in week 3 is easy and in week 10 it is not.

Course learning goals

By the end of this course, the assiduous student will . . .

- . . . set up and solve the equations of motion for a mechanical system in whatever coordinates the problem wants – Cartesian, polar, or generalized – starting from Newton's laws or from a Lagrangian;
- . . . use the conservation laws (momentum, angular momentum, energy) as *tools*, and know when each one applies and why;
- . . . analyze oscillations: damped, driven, resonant, coupled, and their normal modes;
- . . . solve the central-force problem and read an orbit off an effective potential; work in accelerating and rotating frames;
- . . . explain what chaos is and is not, and recognize it in a simulation;
- . . . use Python to solve equations of motion numerically, to check analytic results, and to explore what happens when the algebra gives out;
- . . . and check their work the way physicists do: units, limiting cases, symmetry, and plugging the answer back in.

Calendar

The complete day-by-day schedule (topics, reading due, PCCIs, homework due) is posted on Moodle and kept up to date there. In outline:

Weeks	Topics	Taylor
1	Newton's laws; polar coordinates	Ch. 1
2	Projectiles with air resistance; charged particles in magnetic fields	Ch. 2
3	Momentum and angular momentum; rockets; center of mass	Ch. 3
4	Energy: conservative forces, potential energy, central forces	Ch. 4
5–6	Oscillations: damped, driven, resonance, Fourier series	Ch. 5
7–8	Calculus of variations; Euler–Lagrange; the brachistochrone	Ch. 6
8–9	Lagrangian mechanics: generalized coordinates, constraints, multipliers	Ch. 7
9–10	Two-body central-force problem; orbits	Ch. 8
11	Non-inertial frames: tides, centrifugal and Coriolis forces	Ch. 9
12–13	Coupled oscillators and normal modes; the double pendulum	Ch. 11
14–15	Chaos	Ch. 12

Key dates.	Mon Oct 5	Mountain Day holder
	Mon Oct 12	No class (autumn recess)
	Mon Oct 19	Exam 1 (Ch. 2–4)
	Fri Nov 13	Exam 2 (Ch. 5–7)
	Nov 25, 27	No class (Thanksgiving)
	Mon Dec 7	Exam 3 (Ch. 8, 9, 11)
	Dec 19–22	Final exam period (self-scheduled)

Mountain Day is announced the morning of; if it lands on a class day, that day's material shifts into the Oct 5 holder.

Grading breakdown

Everything in this course is worth points, and the course total is exactly 1000. Within each category, each assignment is weighted equally. Letter grades are not curved.

Category	Number	Drop	Each	Total
Participation & pre-class check-ins (PCCIs)	39	4	2	70
Weekly homework	13	1	25	300
Exams (in class)	3	0	190	570
Final exam, required part	1	0	60	60
Course total				**1000**

Grade scale (percent of 1000; not curved):

A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E
93+	90	87	83	80	77	73	70	67	63	60	< 60

Each cutoff is the bottom of that grade (a B- is 80–82.9, and so on).

Pre-class check-ins and attendance

Most class days there will be a short PCCI due at the start of class, on paper: usually one Taylor problem, chosen so that working it primes you for that day's material. **You will receive full credit for a good-faith effort that includes a clear description of anything you are stuck on.** An incorrect or incomplete answer *without* such an explanation won't receive full credit. For example, if the question were to differentiate $f(g(x))$: "I don't know" earns nothing; the correct answer earns full credit; and so does "I think it's $f'(g(x))$, except I know the chain rule says to do something different because it's $f(g(x))$ and not just x – I'm not clear on what, though." A PCCI should take you less than 15 minutes; if one doesn't, please tell me.

Turning in the day's PCCI is also how attendance is recorded, and your four lowest participation days are dropped, so occasional absences and off days cost you nothing. This is a small class, and it runs on group work at the board; your classmates are a resource for you and vice versa, and that in-class work is one of the main benefits of a school like Smith. If you have an accommodation, please come talk to me so that we can make the class work for both of us, but *in general*, I will expect you to miss no more than four classes, and missing more than four may adversely affect your grade.

Homework

Physics is learned by doing problems, and upper-level mechanics problems are long: expect a set of five or six Taylor problems each week, most of them substantial. Homework is posted on Moodle and due there **Wednesdays by 10:00 PM** as a single PDF of your written work (photos of handwritten pages are fine; combine them into one PDF). A few due dates move to sit before an exam or a break; the Moodle calendar is authoritative. Most weeks one problem is **computational**, done in Python on posit.smith.edu (see *Python* below); include your code and its plots in the PDF.

You may (and should) work with others on the homework, but what you submit must represent your own understanding, written up by you. Your lowest homework score is dropped.

Every problem ends with an assessment. After the answer, write two or three lines checking it: a limiting case ($m \to 0$, $t \to \infty$, the angle going to zero), a units check, or a comparison with a result you already trust. Some problems on the sets say "include an assessment" or "this is your assessment"; the habit is expected on all of them, and it is worth points in the rubric.

Grading methodology for homework (Will Raven's, reproduced from his Fall 2025 syllabus):

- $\checkmark^{+}$ = 100%: The problem, including units and assessment when appropriate, is completely correct.
- $\checkmark$ = 95%: There was a small problem.
- $\checkmark^{-}$ = 50%: There were multiple small problems or a big problem. Submitting a completely correct solution along with a few sentences reflecting on the problem increases your grade up to an 85%.
- X = 25%: There was a lot of confusion on the topic. Submitting a completely correct solution along with an in-depth reflection on the problem increases your grade up to a 75%.
- 0: You skipped the problem.

Your reflection should address why you got the problem wrong, your initial misconceptions, and, most importantly, what you have learned to correct those misconceptions. Research in pedagogy highlights reflection as one of the most effective learning tools. Resubmissions and reflections are due within 10 days after the homework is returned and must be submitted on a separate paper along with the next homework set. Reflections that lack sufficient depth or fail to demonstrate meaningful engagement with the problem will result in fewer points.

Exams

Three exams, in class, taking the full period, on the dates above. Each exam has **three problems, one per chapter covered**; a student with command of the material should finish each in about 25 minutes. Exams are closed-book; you may bring **one 8.5×11 sheet of handwritten notes, both sides**, and a basic calculator (no phones, laptops, or other internet-connected devices). Students with documented accommodations will, of course, have those accommodations met. Exam problems are closely aligned with the in-class problems and the homework, and each exam covers only the chapters listed – never the week the exam is given.

The final exam, during the December 19–22 self-scheduled exam period, has two parts:

1. **Required** (60 points): one problem on Chapter 12 (chaos), the only chapter not covered by an in-class exam.
2. **Optional grade-replacement problems**: nine more problems, one for each chapter that appeared on the three exams (Chapters 2–9 and 11). Attempt any of them. For each one you solve, the new score *replaces* your score on the corresponding exam problem if and only if it is higher. There is no penalty for an attempt; this can only improve your grade.

Do the required problem first; then do as many redemption problems as you want and have time for. I keep per-problem exam scores all semester precisely so this works.

Partial credit, and how to get it. Several of the problems in this class are quite challenging. For the harder problems, my goal is to have you make the strongest possible effort toward **understanding** the solution. Thus, if you cannot fully solve a problem, say whatever you can about the way in which a solution would proceed from where you stop; say whatever you can about the qualitative behavior of a solution; say whatever you can about the physical meaning of the solution. And always, always: **check your units**, **check the limiting cases**, and **plug your solution back into the equation of motion**. You are never done solving an equation of motion until you have verified that your solution works.

Late policy

It is extremely hard to catch up on late work (usually things are late because you are busy, and people don't tend to get less busy!). That is why every category has built-in drops: four participation days, one full homework set. My strong suggestion, when life happens, is to use the drops and prioritize getting your schedule back under control.

Beyond the drops: all students start the term with **3 free late passes**. You can spend a late pass to extend a homework set to the following class period with no penalty. Late passes are not automatic – email me before the deadline to say you're using one, and once declared it is spent whether or not you turn the work in by the extended date. Late passes cannot be applied to PCCIs (their whole value is priming you for that day's class – the drops cover them), to exams, or to the final. After your free passes are spent, further extensions cost 10%, then 20%, and so on, of that assignment's points.

Python (and Mathematica)

Computation is part of doing mechanics: the equations of motion for quadratic drag, the driven pendulum, and the double pendulum have no closed-form solutions, and numerical solutions are how physicists actually see those systems. We will use Python (NumPy, SciPy, SymPy, Matplotlib) in Jupyter notebooks on https://posit.smith.edu, the same setup as PHY 210. I'll post starter notebooks for the computational homework problems and for the in-class simulations. Previous offerings used Mathematica, and

Smith has a license; if you already know it and prefer it, you may use it for any computational problem.

Two expectations, borrowed from Will Raven's Mathematica policy for this course:

1. **Simplification first.** For homework and exams, anything that can be done with physics – an integral that vanishes by symmetry or because the integrand is odd, a limit you can read off – gets done on paper first. The computer confirms; it does not replace your understanding of the core principles.

2. **Exploration and assessment.** Use the computer to push past the algebra: vary a parameter, plot the thing, check a limiting case numerically. Say in your write-up what you predicted *before* running it and what you concluded *after*.

Example homework problem using Python (the problem and the approach are Will's; the code is the Python version of what he did in Mathematica). *Consider an infinitely long, uniform rod of mass μ per unit length that lies along the z-axis. Calculate the gravitational force on a point mass m at a distance ρ from the z-axis.*

Solution. First I consider the force on m due to a short segment dz of the rod at height z above m. It has magnitude $dF = Gm\,\mu\,dz/r^2$, directed toward the segment, where $r = \sqrt{z^2+\rho^2}$. To find the total force I need to integrate from $z=-\infty$ to $+\infty$. When I do, the z-components cancel by symmetry (the segment at $-z$ pulls up exactly as hard as the one at z pulls down), and the component perpendicular to the plane of the rod and the mass vanishes by cylindrical symmetry. So I only need the radial component, $dF_\rho = -dF\cos\alpha = -Gm\mu\,\rho\,dz/r^3$, with the minus sign because it points toward the rod. That leaves

$$F_\rho = -Gm\mu\rho\int_{-\infty}^{\infty}\frac{dz}{(z^2+\rho^2)^{3/2}}.$$

I used Python for the last step – SymPy for the integral itself, and then, just for fun, I had it confirm that the z-component really is zero (it must be: the integrand is odd):

```
import sympy as sp
G, m, mu, rho = sp.symbols("G m mu rho", positive=True)
z = sp.symbols("z", real=True)
r = sp.sqrt(z**2 + rho**2)
F_rho = sp.integrate(-G*m*mu*rho / r**3, (z, -sp.oo, sp.oo))
F_z   = sp.integrate(-G*m*mu*z   / r**3, (z, -sp.oo, sp.oo))
print(F_rho, F_z)     # prints:  -2*G*m*mu/rho   0
```

So the final answer is

$$\vec{F} = -\frac{2Gm\mu}{\rho}\,\hat{\rho}.$$

Assessment: the units work ($G\mu m/\rho$ has the units of GMm/r^2 with $M=\mu\rho$), and the $1/\rho$ falloff (rather than $1/\rho^2$) is what an infinite line source should give – the same thing happens for the electric field of an infinite line charge in PHY 118. While I plugged the second integral into SymPy, it must be zero since the integrand is odd; the computer is confirming the symmetry argument, not replacing it.

Inclusion and equity

I can hardly overstate how much inclusion, equity and diversity considerations have shaped my entire thinking around teaching and practicing physics. It affects everything from the way that physics is framed historically to the way that it is typically taught in your classrooms. My main goal is to create an environment that tells all of us that we can do physics; that invites people into physics who have never been invited, or who have been explicitly excluded.

Physics cannot be divorced from the world in which we study it. I expect that we will recognize that in this class, and I expect that we will create an inclusive and welcoming environment. More specifically, I expect that we will strive *not* to create an environment that excludes people. This is hard work. We will get things wrong. You will get things wrong. *I* will get things wrong. Especially when I get things wrong, I encourage you to tell me, using any of the channels under *Sensitive issues* below.

Use of artificial intelligence

This statement, and the prompts that follow it, are Will Raven's, reproduced from his Fall 2025 syllabus for this course with my thanks; I have changed only the name of Smith's integrity policy and added Python where he wrote Mathematica.

I find AI to be a pretty cool and powerful tool, and it offers incredible potential to enhance your learning experience in this course. However, like any powerful tool, it needs to be used wisely. Used improperly, it can genuinely hinder your development as a physicist.

You are welcome to use AI in this class, provided you always adhere to the Academic Integrity Statement (below). Crucially, **using AI to directly solve homework or exam problems is a clear violation of the Academic Integrity Statement and will be treated as such.** Your goal here is to learn, to grapple with challenging concepts, and to develop your own problem-solving abilities. AI can be a fantastic accelerator for that learning, but it cannot replace the essential process of your own intellectual struggle and discovery.

Be aware that AI models, while impressive, are not infallible. They can (and often do) make subtle errors, "hallucinate" incorrect information, or provide solutions that, while technically correct, don't foster the deeper understanding you need. Always approach AI-generated content with a critical eye, just as you would any other resource. Your job is to verify, question, and ultimately understand the physics.

I strongly encourage you to experiment with AI as a study partner. It can be a powerful resource for clarifying concepts, debugging code, or practicing your critical thinking. Think of it as a sophisticated tutor available 24/7. Below are some recommended ways to start using AI effectively. (Two additions of mine: PCCIs are graded on good-faith effort, so the only thing being graded is a record of *your* thinking, and having AI polish that helps your grade not at all; and early in the semester we will spend part of a class deciding *together* how AI should and shouldn't fit into homework in this course, computational work especially, and post what we decide on Moodle next to this policy.)

Recommended AI prompts for learning. When using AI, it's helpful to give it context. Always start by telling the AI who you are and what you're studying.

Initial context-setting prompt (always include this first): "I am a physics major taking upper-level classical mechanics at Smith College. I am currently studying [specific topic, e.g., Lagrangian mechanics, coupled oscillators]. Please tailor your responses to this context, assuming I have a weak/medium/strong foundation in calculus, differential equations, and introductory physics."

1. **For conceptual understanding and definitions (ask the AI to ask *you* questions).** Use this when you want to solidify your grasp of a concept or definition. Instead of just getting an answer, prompt the AI to help you critically evaluate your own understanding. "I'm trying to deepen my understanding of [specific topic, e.g., time-dependent potential energy] in classical mechanics. Here's my current understanding/definition: [your definition]. Instead of just giving me a refined definition, can you ask me a series of probing questions that would help me critically evaluate and improve my understanding, focusing on the fundamental implications and common misconceptions related to [topic] and its role in classical mechanics?"

2. **For programming help (focus on debugging skills, not just fixes).** When your code isn't working, use AI to learn how to debug systematically. "I'm working on a [Python, Mathematica] program to model [what it does, e.g., the motion of a 2D projectile subject to quadratic drag], and I'm encountering issues. My goal is not just to fix the code, but to understand the debugging process. Here is my program: [code]. Can you help me identify potential errors by explaining your diagnostic process rather than just providing the corrected code? Focus on common pitfalls in physics simulations and how to systematically approach debugging them. If there are multiple errors, address them one at a time."
3. **For refining reflections and critical thinking.** This is excellent for analyzing why you made mistakes and how to avoid them in the future. "I'm preparing a reflection for a classical mechanics problem where I struggled. The purpose of this reflection is to critically analyze my mistakes and articulate what I've learned to prevent similar errors in the future, thereby strengthening my problem-solving and critical thinking skills as a physicist. Here's the problem: [problem]. My primary point of confusion or error was: [specific confusion/error]. Can you act as a mentor and ask me a series of thought-provoking questions that will guide me in drafting a comprehensive and insightful reflection? Focus on probing the *conceptual* rather than just computational errors, and push me to think about the underlying physics principles and problem-solving strategies."
4. **For conversational exploration of confusing topics.** When a topic just isn't clicking, use AI as a dialogue partner to explore it in depth. "I want to have a detailed, high-level conversation about [confusing topic, e.g., time-dependent potentials] in classical mechanics. My goal is to develop a robust conceptual and mathematical understanding. Can you start by explaining a key fundamental aspect or a common misconception, and then we can explore it through a dialogue? Feel free to challenge my assumptions or prompt me with scenarios."
5. **"What if?" scenarios (predict and explore variations).** Challenge yourself by exploring how changes to a problem affect the outcome. "I've just solved a classical mechanics problem involving [e.g., a simple pendulum with damping]. To deepen my understanding, can you propose a 'what if' scenario that slightly modifies the problem parameters or introduces a new element, and then ask me to predict the qualitative (and perhaps quantitative) impact? We can then discuss the implications and verify my reasoning. The goal is to explore the boundaries of my understanding of [relevant physical principle]."

The physics department's longer "AI prompts for students" document, with role-specific prompts (the code debugger, the math coach, the exam study buddy), is posted on Moodle.

COMMUNICATION

My preferred mode of communication is email. I will check my email multiple times per day during the work week. If you have sent me an email and have not heard back from me within 24 hours (excluding weekends), you should email me again or drop by my office to ask your question in person. I cannot guarantee that I will regularly check my email on evenings or on weekends.

I expect that you will check your Smith email at least once per day: 24 hours after I email the class, I expect everyone has read it. Emailed assignments, due dates, and policy changes carry the same weight as this syllabus, and the same goes for Moodle.

Workload. This course should take about 12 hours of your time per week: about 4 in class, and about 8 outside it on reading, PCCIs, homework problems, and review. **If you find yourself spending significantly more (or less) than this, please let me know ASAP! That's an instructor-error that I can easily fix.**

Sensitive issues What if you have something you want to talk about, but don't feel comfortable doing so in a public (or non-anonymous) space? This often comes up with issues of racism, sexism, and other kinds of discrimination. It can come up with issues that relate to my behavior, the behavior of your classmates, or the world in general. Please feel *encouraged* to use either the anonymous feedback form linked on Moodle, or to leave a written note under my door (if you want to write a note, you may want to type it up and print it out so that your handwriting is further anonymized). Of course, if you feel comfortable and safe doing so, you are also encouraged to talk publicly or non-anonymously about these things! If you are not comfortable with any of these options, the *Getting help* section below lists offices that can help, including confidential ones.

Recording My plan is to record from my iPad to both PDF and videos, posted to Moodle. We'll see how technology works. I do this because several students have used it as a post-class resource in the past. Sometimes a derivation is worth reviewing, sometimes you were having a spacey day, etc. But it's worth knowing that student experience shows that it's a very poor substitute for attendance, and just watching the recordings won't count towards your 70 attendance/participation points. Sometimes the mic picks up conversations in the room. If that's a problem, I can look into muting it. Let me know.

Academic integrity Real learning happens when students conduct themselves with academic integrity, and our goal here is to provide you with the opportunity for (and expectation of) that learning. Read the Academic Integrity Statement. That statement is a gift to the community, giving you all sorts of privileges (self-scheduled finals, the expectation that I treat you like academic adults, etc.). And it comes with responsibilities. All submitted work of any kind must be the original work of the student, who must cite all the sources used in its preparation – including, for example, an AI model.

In this class specifically: you are encouraged to work together on the homework and to study together for exams, but PCCIs should be attempted individually first, everything you turn in must be your own work, and exams and the final must be done individually. If you cannot explain why you have written what is on your paper, you have crossed from collaborating to copying.

Academic accommodations If you need classroom accommodations, please contact the Accessibility Resource Center and file things through the Accommodate system. I promise to be flexible in meeting any documented need for accommodations in a way that works well for you. If you would like some accommodation but do not have an ARC reference, let me know and we can discuss your concerns: regardless, I want to provide you the support you need to succeed in this course, though I will do so in a way that does not diminish the accommodations of others. Remember to contact me well in advance of when you need the accommodations in place, so that we can prepare.

Course feedback There will be an anonymous mid-semester feedback form on Moodle around fall break, and during the last week of classes approximately 15 minutes at the beginning of one class will be set aside for you to complete the official course feedback questionnaire. Please don't wait until then if something isn't working.

Getting help **Books.** Taylor is unusually readable; most weeks the reading is the best preparation you can do. When you want a second voice: Morin, *Introduction to Classical Mechanics*, has the best problem collection in the subject (with solutions) and a more demanding style; Kleppner & Kolenkow, *An Introduction to Mechanics*, is the classic at a slightly lower level and is excellent on rotating frames and angular momentum; Marion & Thornton, *Classical Dynamics of Particles and Systems*, is the traditional alternative to Taylor and covers the same ground in a different order. Feynman's lecture on the principle of least action (Volume II, Chapter 19, free at feynmanlectures.caltech.edu) is worth reading once we reach Lagrangians. Later in your career: Goldstein, *Classical Mechanics*, the graduate standard.

People and places. There are many resources at Smith to help you succeed in this course and in your college career:

- Your instructor! Think of my office hours as **free help sessions**: if you are going to spend a couple of hours a week on homework anyway, why not do some of it where the person who writes your exams can help? That's my job, so don't think it's bothering me to have you reach out with questions or concerns.
- The Spinelli Center – open hours, one-on-one appointments, and math review workshops.
- Writing help, learning-specialist services, and workshops on time management and managing stress, all from the Jacobson Center.
- Crisis resources, counseling, and wellness services at the Schacht Center.
- Resources for gender identity and expression, and the Office for Equity and Inclusion in general.
- Where to report sexual misconduct and other discrimination. Please note that your instructors are Responsible Reporters.